



Predicting Wine Quality:

A Comparison of Linear Regression Techniques on a Multicollinear and Outlier-Affected Dataset

Report submitted to :

SISTER NIVEDITA UNIVERSITY

For the award of the degree

of

Bachelor of Science (Hons.) Statistics

by

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B.Sc. FINAL YEAR PROJECT COMPLETION CERTIFICATE

This is to certify that **SAMYABRATA ROY** (REGISTRATION NO: **220010093372**) has prepared the project work entitled **'PREDICTING WINE QUALITY: A COMPARISON OF LINEAR REGRESSION TECHNIQUES ON A MULTICOLLINEAR AND OUTLIER-AFFECTED DATASET'** under the supervision of **PROF. ANIRUDDHA CHOUDHURI** based on the survey of literature in his/her area of interest for the partial fulfilment of the B.Sc. (Hons.) Degree in Statistics from Sister Nivedita University.

Dr. Anindita Ghosal

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Signature of External with date

Signature of Supervisor with date



DECLARATION BY THE STUDENT

I hereby declare that the work which is being presented in this dissertation entitled “Predicting Wine Quality: A Comparison of Linear Regression Techniques on a Multicollinear and Outlier-Affected Dataset” in partial fulfilment of the requirements for the award of the degree of Bachelors of Science in Statistics is an authentic record of my own work carried out during the period from 6th January 2025 to 30th April 2025 under the supervision of Prof. Aniruddha Choudhuri, Department of Statistics, Sister Nivedita University.

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I further declare that no portion of the dissertation or its data will be published without the Institute’s or supervisor’s permission. I have not previously applied for any other degree or award using the topics and findings described in my dissertation.

(Signature of the students with Date)

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Acknowledgement

With immense pleasure and a deep sense of gratitude, I wish to express my sincere thanks to my supervisor, **Prof. Aniruddha Choudhuri**, Assistant Professor, Department of Statistics, Sister Nivedita University (SNU), for his motivation and continuous encouragement. Without his support, this project would not have been successfully completed.

I am grateful to the Chancellor of SNU, **Dr. Satyam Roy Choudhuri**, Vice Chancellor **Prof. (Dr.) Dhrubojyoti Chattopadhyay** and the Registrar, **Prof. Suman Chatterjee** of SNU, for motivating me to carry out this project at their university.

I express my sincere thanks to **Prof. (Dr.) Mala Mitra**, Dean, School of Sciences, and **Dr. Anindita Ghosal**, Head, Department of Statistics, School of Sciences, Sister Nivedita University, for their continuous support and encouragement.

I would like to convey a special note of thanks to **Prof. Argha Nath Bhattacharya**, Department of Statistics, for his insightful suggestions and timely feedback, which played a crucial role in shaping the direction of this project. His approachability and willingness to help truly made a difference.

I also wish to extend my profound sense of gratitude to my parents for providing me with the moral support and encouragement whenever required.

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Date: 02/05/2025

Contents

Particulars	Page No.
1. Certificate	2
2. Declaration by the Student	3
3. Acknowledgement	4
4. Abstract	6
5. Introduction	7
6. Objectives	9
7. Data Collection	10
8. Data Description	10
9. Methodologies	12
9.1 Summary of the Dataset	12
9.2 Exploratory Data Analysis (EDA)	13
9.3 Regression Analysis	19
10. Model Fitting	31
11. Model Comparison	31
12. Cross-validation	32
13. Conclusions	33
14. Limitations and Future Scope	34
15. References	35
16. Appendix	37
16.1. Dataset	37
16.2. R Code	39
17. Plagiarism Report	46

Abstract

This project explores the application of various linear regression techniques for predicting wine quality, which has a very important role, particularly in the wine industry for shaping up the preferences of consumers, influencing the pricing related strategies and guiding decision making in production, using a dataset characterised by multicollinearity and a significant presence of outliers. The dataset comprises 32 wine samples, each evaluated across 10 physicochemical attributes - including pH, total sulphur dioxide, anthocyanin concentration and colour density - with quality scores assigned on a 0-20 scale. An extensive Exploratory Data Analysis (EDA) revealed distributional skewness, strong inter-variable correlations, and a high incidence of outliers. The initial application of Ordinary Least Squares (OLS) regression exposed limitations due to multicollinearity, as confirmed through elevated Variance Inflation Factors (VIFs). Stepwise regression (both-way) improved model parsimony, but heteroscedasticity and sensitivity to outliers persisted. To address these, robust regression approaches were adopted, including Huber's M-estimator and the MM-estimator. Comparative analysis using metrics such as Adjusted R^2 and Root Mean Square Error (RMSE) demonstrated the MM-estimator's superior resilience to data irregularities. Ultimately, the MM-estimator emerged as the most reliable and interpretable model, offering a robust framework for data-driven wine quality assessment and decision-making in viticulture. This project thus underscores the importance of robust techniques in real-world data environments and presents a generalisable modelling framework to support objective wine quality assessment in the wine industry.

Keywords: Wine quality rating; Exploratory Data Analysis; Ordinary Least Squares; Stepwise Regression; Robust Regression

Introduction

Wine quality plays a crucial role in the wine industry, shaping consumer preferences, influencing pricing strategies, and guiding production decisions. Traditionally, quality assessment has relied on sensory evaluations conducted by expert sommeliers. While valuable, such evaluations can be subjective and prone to inconsistency. The perceived quality of wine is influenced by various physicochemical properties, including acidity, sulfur dioxide content, colour density, and anthocyanin composition. Understanding the quantitative relationship between these measurable features and wine quality allows for a more objective and systematic evaluation process, reducing dependence on human judgment. With the rise of data-driven approaches in oenology, there is growing interest in predictive modelling techniques that assess wine quality based on these measurable characteristics.

Numerous projects have explored the feasibility of using statistical and machine learning methods for wine quality prediction. For instance, Dahal et al. (2021) evaluated models such as Ridge Regression, Support Vector Machines, and Gradient Boosting, demonstrating improved performance over traditional techniques, especially in handling high-dimensional data [\[1\]](#). Similarly, Arshad (2024) employed ensemble models like Random Forest and XGBoost, reporting over 95% accuracy on larger, well-curated datasets [\[2\]](#). However, these projects often overlook the effects of multicollinearity and outliers—issues that can significantly degrade model performance in practical settings.

Addressing this gap, the present project investigates a modest yet noisy dataset comprising 32 wine samples, each described by 10 physicochemical variables, including pH, total sulphur dioxide, anthocyanin concentration, and colour density. The response variable is a wine quality score on a 0–20 scale, derived from sensory assessments. The dataset presents two key challenges: multicollinearity among predictors and the presence of outliers, necessitating the adoption of robust, adaptive modelling strategies suited to real-world data.

Before applying regression techniques, a comprehensive Exploratory Data Analysis (EDA) is conducted to understand the structure and variability of the data. Techniques such as histograms and density plots are used to assess the skewness, modality, and distribution of individual variables, helping to identify those requiring transformation. Boxplots assist in outlier detection, while scatterplots and correlation matrices are used to examine bivariate relationships and detect multicollinearity. Insights from the EDA inform the selection of appropriate modelling techniques, ensuring that the subsequent analysis is both accurate and resilient to data irregularities.

The modelling phase of this project begins with consideration of the foundational assumptions underlying linear regression models—namely, linearity, independence of residuals, and homoscedasticity. Although these assumptions are not formally tested prior to modelling, they are assessed contextually using insights derived from EDA and residual diagnostics throughout the modelling process. The initial modelling approach employs the Ordinary Least Squares (OLS) method, providing a baseline for performance and interpretability. However, the presence

of multicollinearity among predictors could necessitate the exploration of variable selection techniques to improve model robustness.

To mitigate multicollinearity, we might apply stepwise regression techniques, including Forward Selection, Backward Elimination, and Bidirectional Stepwise Selection. These methods iteratively add or remove predictors based on criteria such as the Akaike Information Criterion (AIC), allowing the model to retain only those variables that meaningfully contribute to predictive accuracy. Despite the common use of Ridge Regression to handle multicollinearity, this technique is excluded from consideration due to its sensitivity to outliers. Ridge Regression minimises the sum of squared residuals augmented by an L2 penalty term $(\lambda \sum_{j=1}^p \beta_j^2)$, which disproportionately penalises large coefficients. In the presence of extreme data points, this can distort model estimates and reduce generalizability.

With over 25% of the data points in our small dataset identified as outliers - an insight revealed during EDA - it is neither statistically advisable nor feasible to remove them. Instead, the project adopts and derives the best robust regression techniques designed to accommodate such irregularities without compromising model integrity. These methods offer enhanced resistance to the influence of outliers, ensuring that model estimates remain stable and reliable even under data contamination.

By evaluating the predictive accuracy and interpretability of different regression approaches, this project seeks to identify a robust and generalizable model for wine quality prediction. The findings offer practical insights into the key physicochemical factors influencing perceived quality, contributing both to data-driven winemaking practices and the broader methodological discourse in predictive modelling.

The industrial significance of this project lies in its potential to support quality assurance and product development within the wine industry. Wine quality assessment, often reliant on expert tasters, can benefit from objective, data-driven models that standardise evaluation and reduce subjectivity. By identifying the key physicochemical attributes that influence wine quality, producers can fine-tune cultivation practices, fermentation processes, and blending strategies to consistently meet consumer expectations. Furthermore, predictive models can streamline quality control workflows, reduce production costs, and enhance competitiveness in global markets where quality differentiation plays a pivotal role in brand positioning.

Objectives

> Understand Influencing Factors:

This project aims to uncover how various physicochemical properties, such as acidity, sugar content, pH levels, and alcohol concentration, influence the overall quality of wine. By identifying the most impactful features, the project will help highlight which aspects of wine production require more attention to improve quality outcomes.

- **EDA:**

In the EDA phase, we will perform a comprehensive analysis using various visual and statistical techniques to understand the distribution, variability, and relationships among the variables. **Scatter plots** will be used to reveal potential linear and non-linear relationships between predictors and the target variable (wine quality), while **histograms** will examine the distribution and skewness of individual features. **Box plots** will help to detect outliers and compare feature distributions across different wine quality levels. Generating a **correlation matrix** will help to identify multicollinearity among features, highlighting which variables are strongly interrelated. Finally, the **Coefficient of Variation (CV)** will be calculated for each feature to assess their relative variability and importance in influencing wine quality. These insights will lay the foundation for feature selection and modelling decisions.

- **Build and Analyse Predictive Models:**

The project will employ multiple regression-based linear modelling techniques to predict wine quality scores with a focus on accuracy, interpretability, and robustness. Techniques such as **Ordinary Least Squares (OLS) regression**, **Stepwise regression**, **Regularisation Techniques** and **Robust regression (using Huber Loss, MM-estimator, etc.)** will be explored as per the requirement of the analysis. These methods will allow for a comprehensive comparison of how different modelling strategies handle issues like multicollinearity, overfitting, the presence of outliers and others. The role of regularisation and variable selection will be particularly emphasised in enhancing model generalizability and stability under real-world data conditions.

- **Evaluate Model Performance:**

To ensure reliability, the models will be evaluated using performance metrics such as Root Mean Squared Error (RMSE) and adjusted R-squared. These metrics will help assess how well each model fits the training dataset and how effectively it generalises to unseen wine data. This dual evaluation provides a balanced understanding of both the in-sample performance and the model's predictive power on new data, guiding the selection of the most suitable modelling approach.

- **Support Quality Control in Wine Production:**

By predicting wine quality based on measurable chemical inputs, the study intends to support wine producers in maintaining consistent quality standards. The results could aid in data-driven decision-making in the wine industry.

Data Collection

The dataset used in this study was originally obtained from the UCI Machine Learning Repository, a well-known resource for machine learning datasets. However, the specific dataset has since been archived and is no longer publicly accessible through the repository. The dataset was shared with me by my professor, who had previously downloaded and preserved it for academic purposes.

While no detailed metadata is available due to its archival status, the dataset appears to pertain to physicochemical and colourimetric properties of wines, likely collected to study factors influencing wine quality. Each row in the dataset represents a wine sample, described by 10 independent variables and one target variable indicating the quality score (on a scale up to 20).

Data Description

The dataset consists of multiple independent (might not be orthogonal) variables (predictors) and one dependent variable (target). The target variable is the **wine quality rating**, which is a score on a scale from 0 to 20, where a higher score indicates better-perceived quality. The dataset comprises a sample of 32 wine observations.

Below is a description of all the variables in the dataset:

- y : Quality rating (maximum score = 20)
- x_1 : Wine varietal (0 = Cabernet Sauvignon, 1 = Shiraz)
- x_2 : pH of the wine
- x_3 : Total sulphur dioxide (SO_2) content in parts per million (ppm)
- x_4 : Colour density of the wine
- x_5 : Wine colour (absorbance measure, unitless)
- x_6 : Polymeric pigment colour (indicates stable colour compounds)
- x_7 : Anthocyanin colour (colour contribution from anthocyanins)
- x_8 : Total anthocyanins (g/L), important phenolic compounds responsible for colour
- x_9 : Degree of ionisation of anthocyanins (expressed in per cent)
- x_{10} : Ionised anthocyanins (expressed in per cent)

These features primarily capture the chemical and colourimetric aspects of red wines, which are known to influence quality perception. The dataset is suitable for regression modelling and exploratory data analysis to identify key variables that impact wine quality.

A preview of the dataset (first few rows) is provided below to give an initial look at the structure and values of the variables.¹

	y	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_10
	<dbl>	<int>	<dbl>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<int>	<dbl>
1	19.2	0	3.85	66	9.35	5.65	2.40	3.25	0.33	19	0.065
2	18.3	0	3.73	79	11.15	6.95	3.15	3.80	0.36	21	0.076
3	17.1	0	3.88	73	9.40	5.75	2.10	3.65	0.40	18	0.073
4	17.3	0	3.86	99	12.85	7.70	3.90	3.80	0.35	22	0.076
5	16.8	0	3.98	75	8.55	5.05	2.05	3.00	0.49	12	0.060
6	16.5	0	3.85	61	10.30	6.20	2.50	3.70	0.38	20	0.074

Table 1: Data Head showing the first 6 rows of the dataset

¹ For the whole dataset, refer to the “[Appendix](#)” Chapter.

Methodologies

This project aims to develop a predictive model for estimating wine quality based on physicochemical parameters measured in wine samples. The primary methodological focus is on exploring and comparing various linear regression techniques to determine the most effective modelling approach, informed by diagnostic checks and performance evaluations. The analysis is conducted using R programming, incorporating key packages such as `MASS` for regression modelling, `car` for diagnostic checks, `robustbase` for alternative regression methods, `corrplot` for visualising correlation patterns, and `glmnet` for applying regularisation techniques (if required). These tools are selected to comprehensively evaluate model accuracy, robustness, and generalisability. By identifying the variables most significantly associated with wine quality, this methodological approach contributes to both empirical understanding in academic contexts and practical applications in data-driven winemaking practices.

➤ **Summary of the Dataset :**

Coefficient of Variation: The Coefficient of Variation (CV) is a standardised measure of dispersion that expresses the ratio of the standard deviation to the mean, making it crucial for comparing variability across datasets with different units or scales; we computed CV as a summary to understand relative variability and identify which variables show more inconsistency relative to their magnitude.

Variables	Coefficient of Variation (CV)
y	11.535699
x_1	89.602867
x_2	3.174758
x_3	54.825594
x_4	31.910174
x_5	34.058630
x_6	37.844245
x_7	34.829570
x_8	20.283178
x_9	39.850510
x_{10}	34.829570

Table 2: Coefficient of Variation Table

➤ Exploratory Data Analysis (EDA) :

- i) **Scatter Plots:** These plots help visualise the relationship between each predictor and the response variable, revealing potential linear trends, clusters, or outliers that may influence the model's performance. y vs x_i 's.

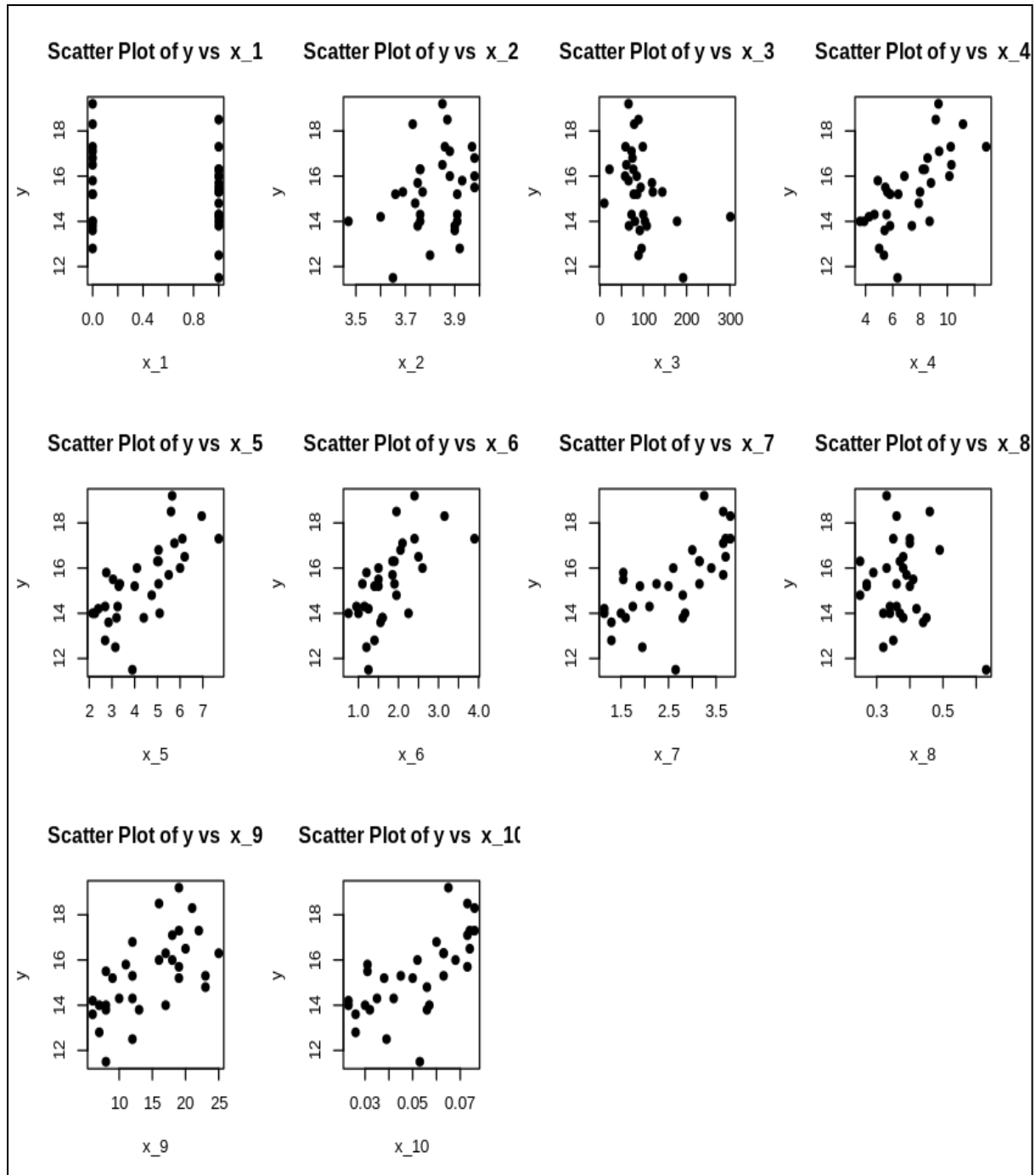


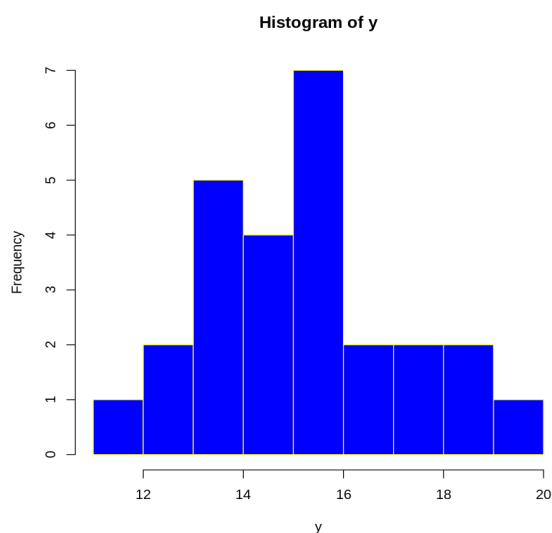
Diagram 1: Scatter plot of y vs x_i 's

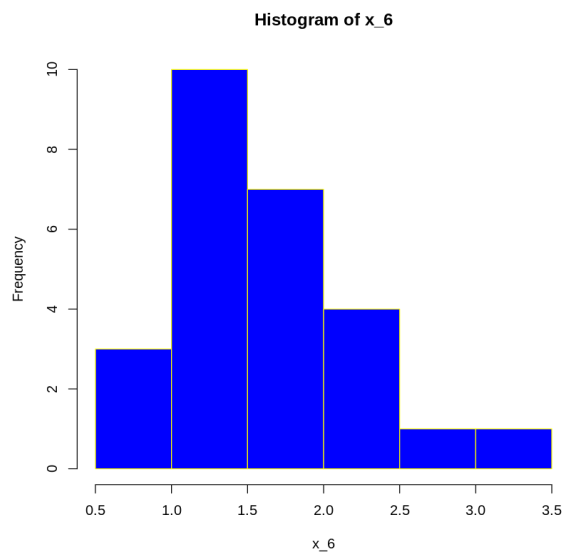
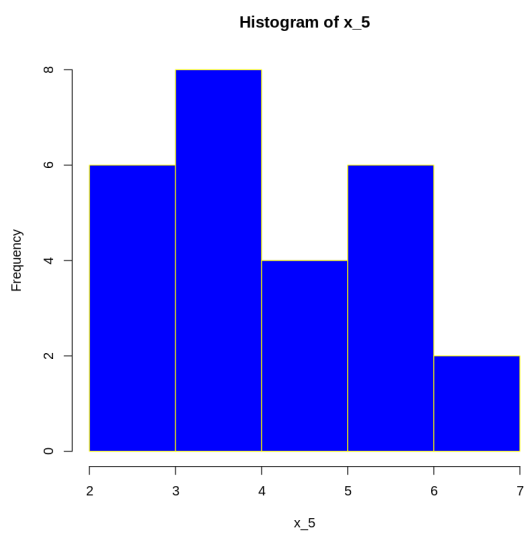
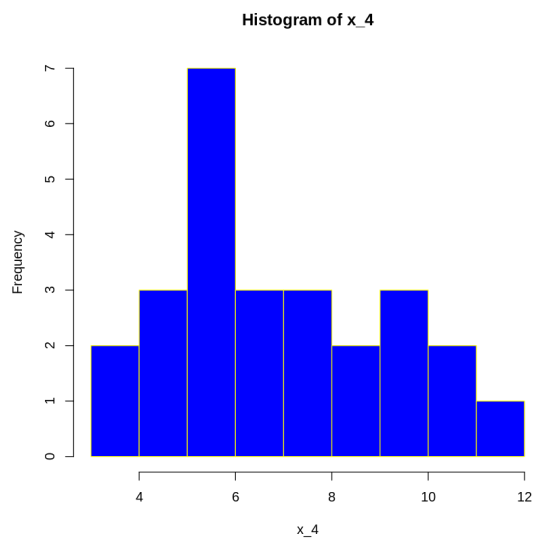
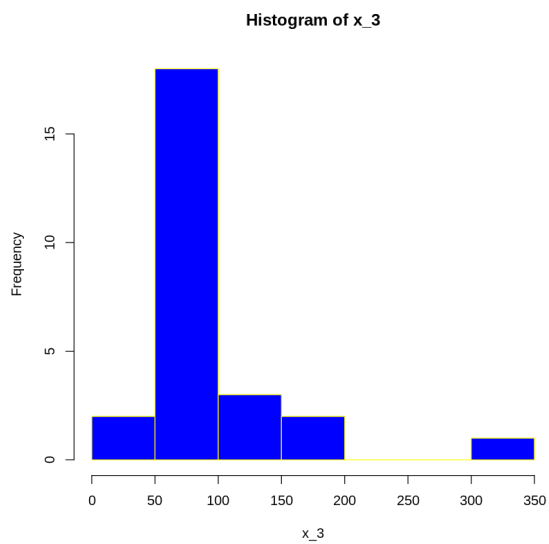
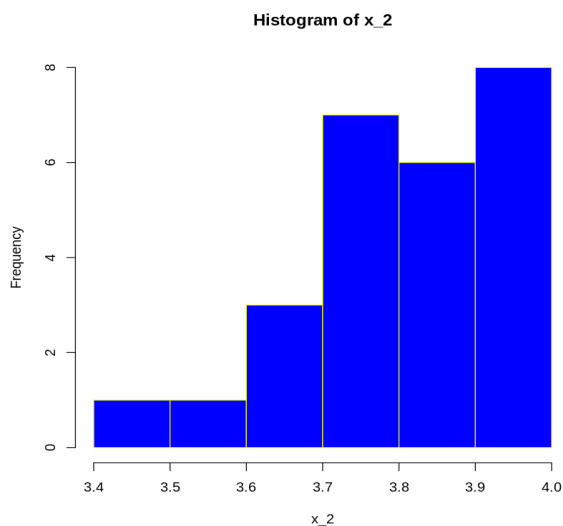
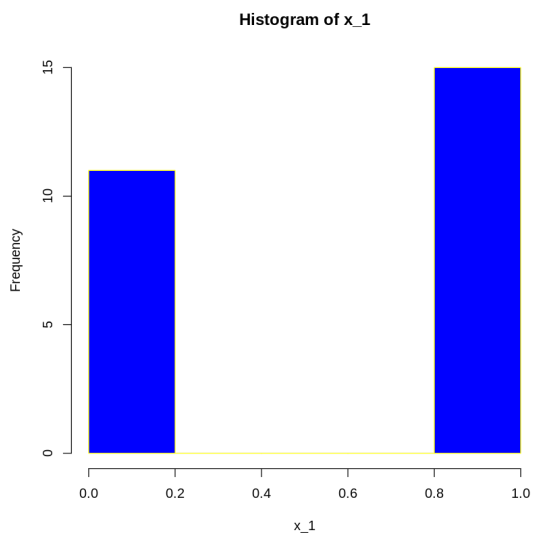
Findings:

Variables	Relationship
y vs x_1	Very little variation in x_1 , making it hard to determine any relationship.
y vs x_2	Moderate scatter, but seems to have a slight positive trend.
y vs x_3	Data appears spread out with no strong correlation.
y vs x_4	Shows a positive linear relationship, though slightly scattered.
y vs x_5	Shows a clear upward trend; positive linear relationship with y .
y vs x_6	Some positive trend, though with more variance, could be moderate.
y vs x_7	Slight positive trend, but weak and with high variance.
y vs x_8	Very scattered, almost no clear trend
y vs x_9	The data is spread and doesn't show a clear pattern.
y vs x_{10}	Slight curvature—might be non-linear or quadratic, but weak overall.

Table 3:

ii) Histograms :





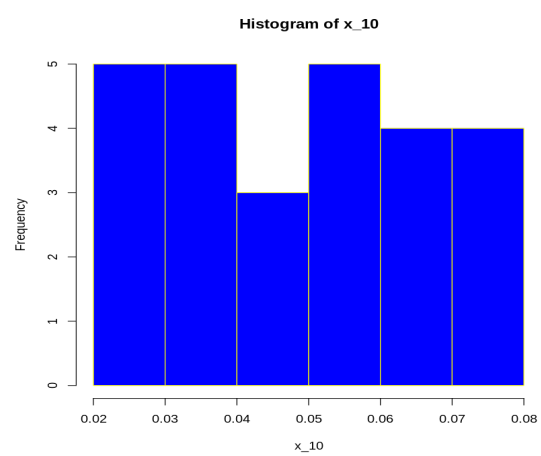
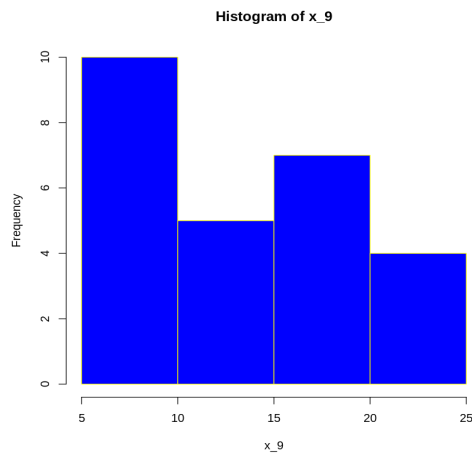
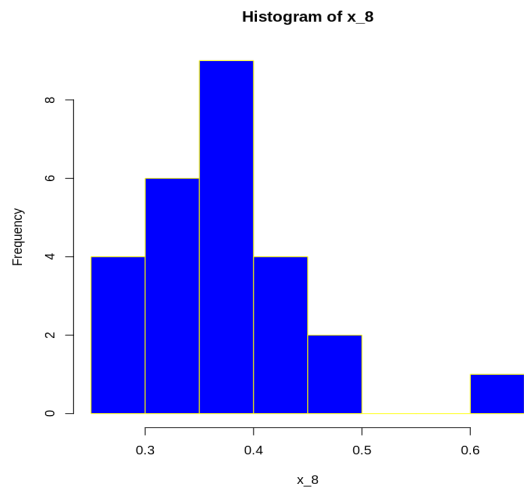
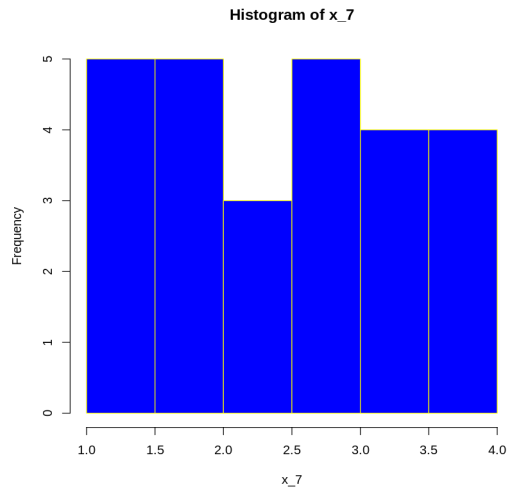


Diagram 2: Histograms

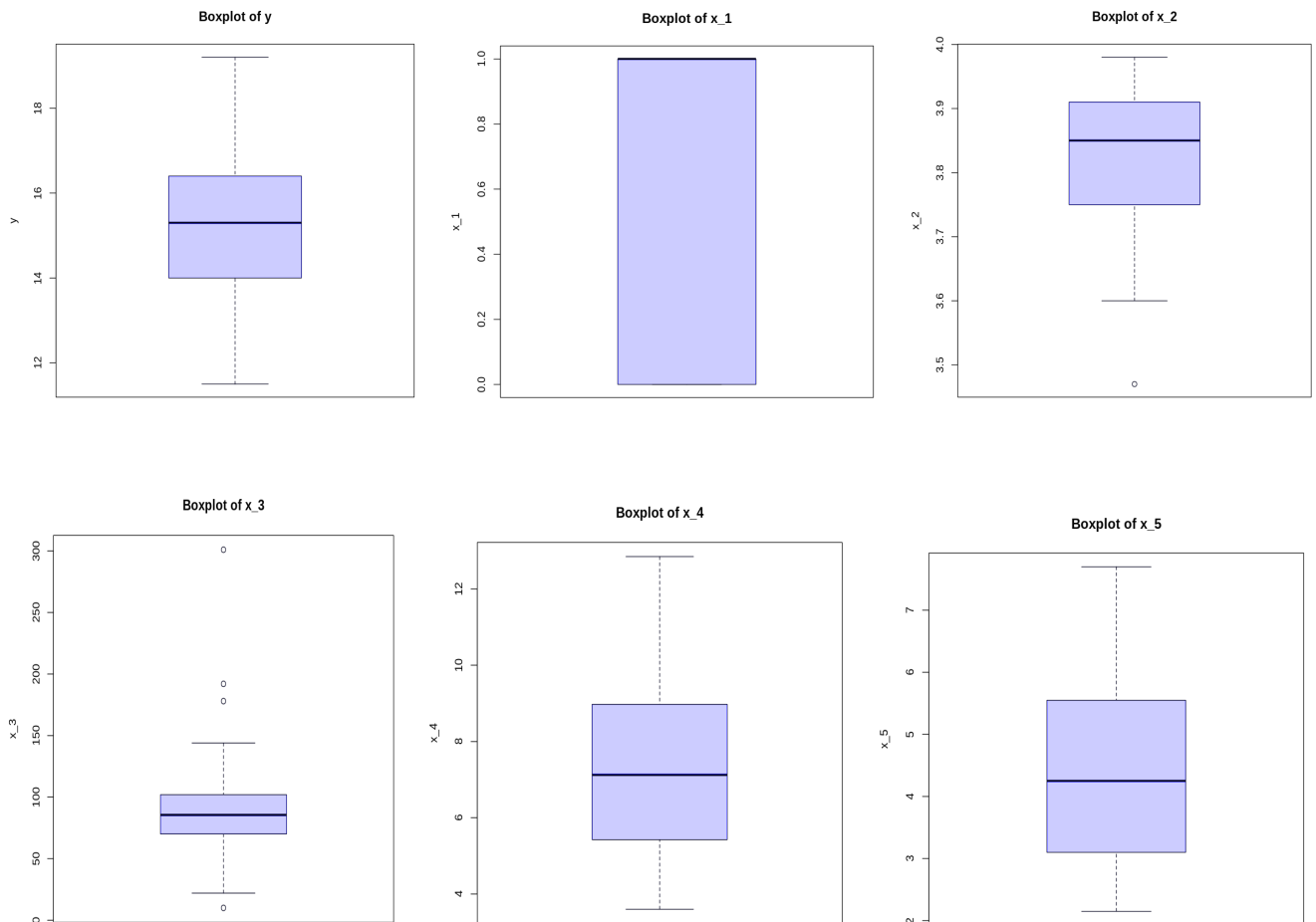
Findings:

Variable	Distribution	Comment
y	approximately normal, slightly right-skewed	Fairly well-distributed; suitable for regression analysis.
x_1	Bimodal / Binary	Categorical variable.
x_2	Left-skewed	Low variability; may have limited predictive power
x_3	Right-skewed	A few large values suggest consideration for transformation if required, while used in regression.
x_4	Fairly uniform, slightly right-skewed	Spread out across a range (around 4 to 12), indicating good variability

x_5	Fairly uniform	It may be a strong predictor.
x_6	Right-skewed	—
x_7	Relatively uniform	The absence of extreme skewness makes it a suitable candidate for modelling
x_8	Moderately right-skewed	The limited range might restrict its explanatory power
x_9	Slight right skew	It has a reasonable spread, indicating potential usefulness
x_{10}	Uniform distribution	Its fairly balanced distribution may still carry predictive value

Table 4: Findings obtained from histograms

iii) Box Plots :



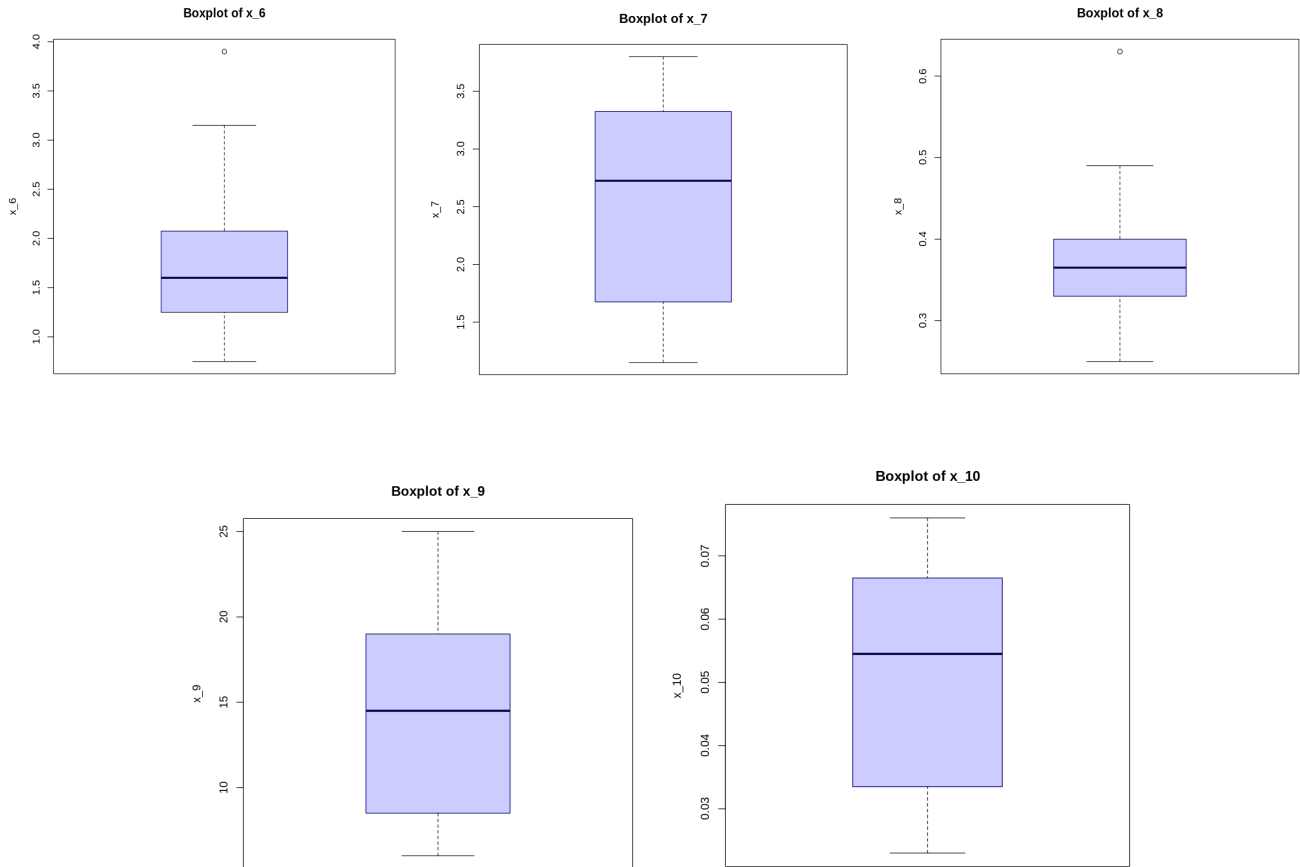


Diagram 3: Box plots

Findings:

Based on the box plots, we observed the presence of outliers in several features. Specifically, x_2 shows an outlier at observation 10, while x_3 exhibits multiple outliers at observations 10, 25, 28 and 32. In the case of x_6 an outlier being detected at observation 4, and x_8 another outlier at observation 32. These outliers could potentially influence the analysis and interpretation, especially in regression-based approaches, and may require further investigation or treatment depending on their impact.

iv) Correlation Matrix: The correlation matrix provides a clear overview of the linear relationships between variables. It helps identify strongly correlated features, which can lead to multicollinearity in regression models. In our analysis, the matrix highlighted pairs of variables with high positive or negative correlations, guiding both feature selection and model refinement by revealing redundancy and potential dependencies among predictors.

	y	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_10
y	1.0000000	-0.169878902	0.27747066	-0.3758899	0.701838522	0.707654712	0.65117305	0.68129132	-0.168180167	0.6170341	0.68129132
x_1	-0.1698789	1.000000000	-0.08880617	0.1150501	-0.008509266	0.012577406	-0.16389528	0.14334447	0.050018542	0.1637408	0.14334447
x_2	0.2774707	-0.088806174	1.00000000	-0.5820282	0.213164035	0.152141791	0.22044010	0.08631043	0.095509243	-0.0489386	0.08631043
x_3	-0.3758899	0.115050136	-0.58202821	1.0000000	-0.391465151	-0.370944023	-0.32542171	-0.36902794	0.404545811	-0.4959928	-0.36902794
x_4	0.7018385	-0.008509266	0.21316403	-0.3914652	1.000000000	0.995666429	0.94541703	0.93671923	0.015519116	0.7968608	0.93671923
x_5	0.7076547	0.012577406	0.15214179	-0.3709440	0.995666429	1.000000000	0.92529036	0.95892682	0.003091977	0.8260006	0.95892682
x_6	0.6511730	-0.163895282	0.22044010	-0.3254217	0.945417026	0.925290358	1.00000000	0.77970743	-0.042853322	0.6905468	0.77970743
x_7	0.6812913	0.143344473	0.08631043	-0.3690279	0.936719226	0.958926821	0.77970743	1.00000000	0.037155356	0.8472281	1.00000000
x_8	-0.1681802	0.050018542	0.09550924	0.4045458	0.015519116	0.003091977	-0.04285332	0.03715536	1.000000000	-0.4558050	0.03715536
x_9	0.6170341	0.163740803	-0.04893860	-0.4959928	0.796860805	0.826000635	0.69054681	0.84722807	-0.455804998	1.0000000	0.84722807
x_10	0.6812913	0.143344473	0.08631043	-0.3690279	0.936719226	0.958926821	0.77970743	1.00000000	0.037155356	0.8472281	1.00000000

Table 5: Correlation Matrix

Findings:

The target variable y shows strong positive correlations with variables such as x_4, x_5, x_6, x_7, x_9, and x_10, indicating that these could serve as effective predictors. However, many of these variables are also highly correlated with one another, raising concerns about multicollinearity, which can affect the stability and interpretability of a regression model. Therefore, careful feature selection or the use of dimensionality reduction techniques is recommended to ensure robust model performance.

➤ Regression Analysis:

We are approaching the wine quality dataset using a linear regression setup, aiming to model and predict the quality score based on various physicochemical features. To ensure reliable model evaluation, the dataset is split into training and testing subsets using an 80:20 ratio through Simple Random Sampling Without Replacement (SRSWOR). This helps maintain randomness while avoiding the repetition of data points in both sets.

i) Ordinary Least Squares Linear Regression:

OLS regression relies on a few key assumptions to ensure valid inference and accurate predictions:

1. Linearity – The relationship between the predictors and the response variable is linear.
2. Normality of Errors – The residuals are normally distributed.
3. Homoscedasticity – The variance of residuals (errors) is constant across all levels of the independent variables.
4. Independence – Errors are independent of each other.
5. No Multicollinearity – Independent variables are not highly correlated with each other.

★ Mathematical Model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p + \varepsilon$$

where,

- y : response variable
- $x_1, x_2, x_3, \dots, x_p$: predictor variables
- β_0 : intercept
- $\beta_1, \beta_2, \beta_3, \dots, \beta_p$: regression coefficients
- ε : random error

Although the [correlation matrix](#) (Table 5) from the Exploratory Data Analysis (EDA) indicates the presence of multicollinearity among several predictors, we still proceed to fit a multiple linear regression model using the Ordinary Least Squares (OLS) method. This decision is made to establish a simple, interpretable baseline against which more advanced or robust models can later be compared. It is important to note that the training dataset, derived from the original data, is expected to inherit similar multicollinearity patterns. Nevertheless, for initial modelling, we assume that the other key assumptions of OLS, namely linearity, homoscedasticity, and independence of residuals, are approximately satisfied. The response variable in this model is the wine quality score, expressed as a linear combination of selected physicochemical attributes. This baseline model provides an essential reference point for evaluating the performance improvements offered by subsequent modelling strategies.

The fitted model (Model: 1.1) provides a summary of the estimated regression coefficients, allowing us to interpret the direction and strength of association between each predictor and the response variable. Which has been shown below:

Residuals:				
Min	1Q	Median	3Q	Max
-1.8138	-0.6850	0.1717	0.4833	1.9544
Coefficients: (2 not defined because of singularities)				
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-9.120845	17.561559	-0.519	0.610
x_1	-0.855781	0.702727	-1.218	0.240
x_2	6.310018	4.263064	1.480	0.157
x_3	0.014215	0.009461	1.502	0.151
x_4	-0.770508	2.593885	-0.297	0.770
x_5	3.386685	3.643216	0.930	0.366
x_6	-1.479912	2.169645	-0.682	0.504
x_7	NA	NA	NA	NA
x_8	-13.450969	8.700101	-1.546	0.140
x_9	-0.125294	0.246570	-0.508	0.618
x_10	NA	NA	NA	NA
Residual standard error: 1.208 on 17 degrees of freedom				
Multiple R-squared: 0.7201, Adjusted R-squared: 0.5883				
F-statistic: 5.466 on 8 and 17 DF, p-value: 0.001612				

Result 1: OLS Model Summary fitted on train dataset (Model - 1.1)

However, the regression summary revealed an important issue—**two coefficients (x_7 and x_{10}) were not estimated**, and the output flagged them as “not defined because of singularities”. This suggests the presence of **perfect multicollinearity**, where some variables

are exact linear combinations of others. As a result, the model cannot uniquely estimate their corresponding coefficients.

To further investigate this, an **alias structure** was examined for the training dataset. Alias detection helps identify **redundant predictors**, which occur when one or more features are **linearly dependent** on others. This dependency causes singularities in the design matrix, making it impossible to estimate unique coefficients for all variables.

The alias output revealed the following dependencies:

Model :										
$y \sim x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10}$										
Complete :										
	(Intercept)	x_1	x_2	x_3	x_4	x_5	x_6	x_8	x_9	
x_7	0	0	0	0	0	1	-1	0	0	
x_10	0	0	0	0	0	1/50	-1/50	0	0	

Result 2: Alias result in the train dataset

This confirms that:

- x_7 is a **perfect linear combination** of x_5 and x_6 , specifically $x_7 = x_5 - x_6$.
- Similarly, x_{10} is also linearly dependent on x_5 and x_6 , i.e. $x_{10} = \frac{x_5 - x_6}{50}$.

Due to these perfect dependencies, the regression model automatically excludes **x_7** and **x_{10}** to avoid multicollinearity.

To address such issues, we will **remove the redundant variables** (x_7 and x_{10}) before fitting the dataset into the OLS model again. The newly fitted model (Model: 1.2) provides a summary of the estimated regression coefficients, allowing us to interpret the direction and strength of association between each predictor and the response variable:

Residuals:				
Min	1Q	Median	3Q	Max
-1.8138	-0.6850	0.1717	0.4833	1.9544
Coefficients:				
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-9.120845	17.561559	-0.519	0.610
x_1	-0.855781	0.702727	-1.218	0.240
x_2	6.310018	4.263064	1.480	0.157
x_3	0.014215	0.009461	1.502	0.151
x_4	-0.770508	2.593885	-0.297	0.770
x_5	3.386685	3.643216	0.930	0.366
x_6	-1.479912	2.169645	-0.682	0.504
x_8	-13.450969	8.700101	-1.546	0.140
x_9	-0.125294	0.246570	-0.508	0.618
Residual standard error: 1.208 on 17 degrees of freedom				
Multiple R-squared: 0.7201, Adjusted R-squared: 0.5883				
F-statistic: 5.466 on 8 and 17 DF, p-value: 0.001612				

Result 3: OLS Model Summary fitted on the train dataset after deleting singularities (Model: 1.2)

This model shows a moderate overall fit with an Adjusted R-squared of 0.5883, indicating that it explains a substantial portion of the variance in the response variable. However, none of the individual predictors are statistically significant, validating the potential issues with multicollinearity. The large standard errors and low adjusted R-squared further support this. Now, due to the indication of high multicollinearity among the predictors, we proceeded to check the Variance Inflation Factor (VIF) scores for the parameters in the model (Model: 1.2).

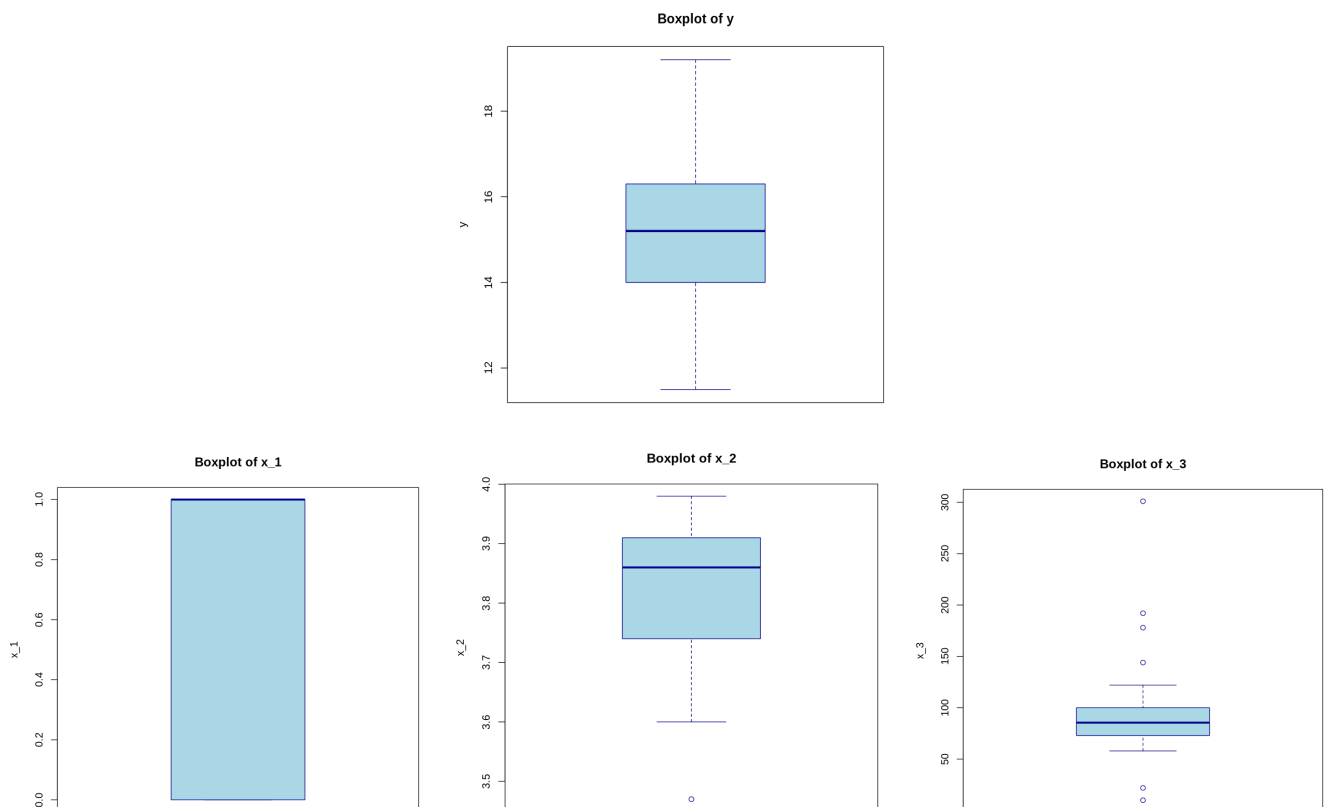
VIF Scores:

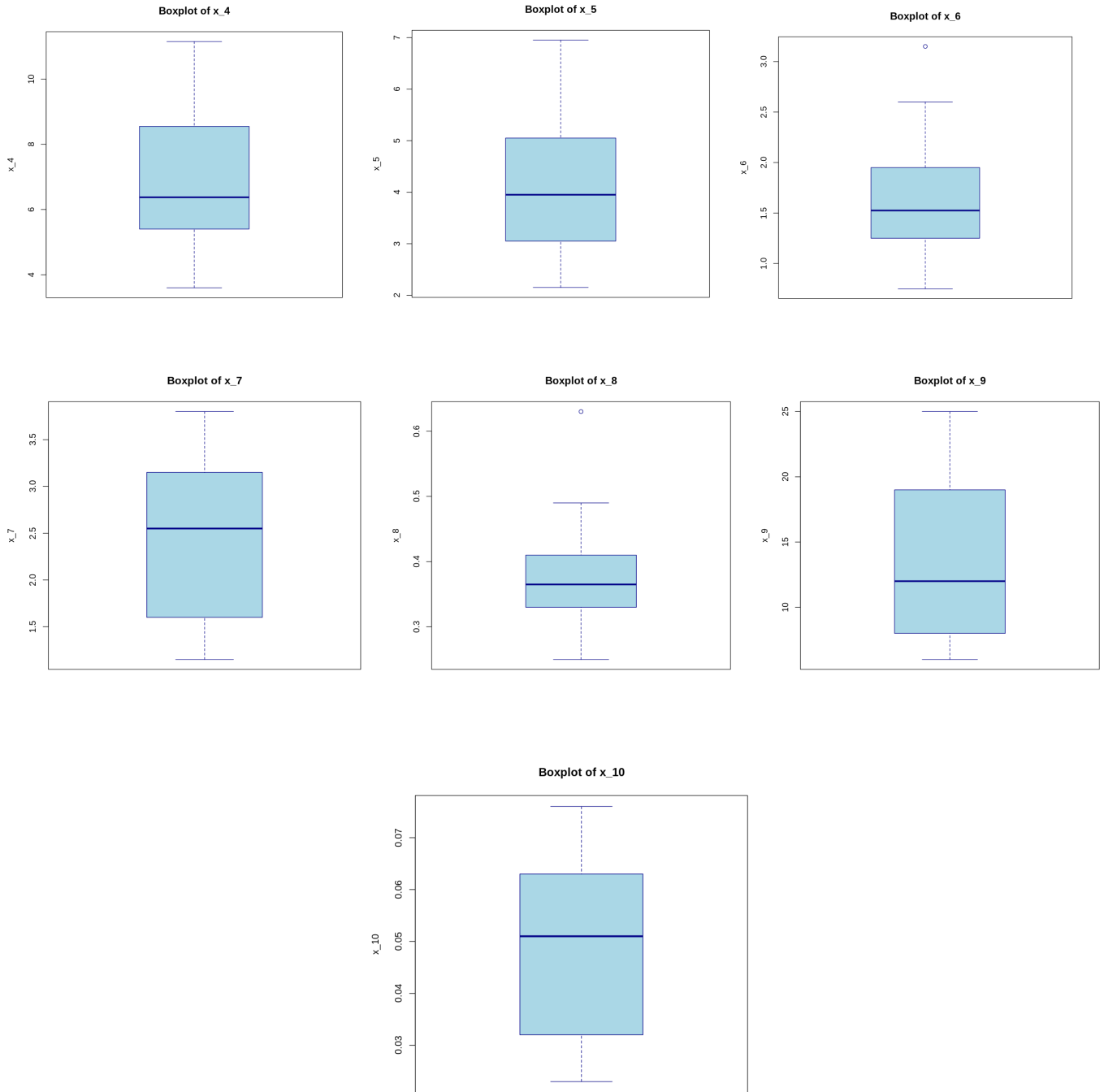
x_1	x_2	x_3	x_4	x_5	x_6	x_8
2.149156	5.339411	4.983590	520.757328	424.374295	25.588846	8.639865
x_9						
36.351670						

Result 4: VIF scores from Model 1.2

The VIF results confirm the presence of severe multicollinearity in the model. Several predictors, notably x_4 (VIF ≈ 520.76), x_5 (VIF ≈ 424.37), and x_9 (VIF ≈ 36.35), exhibit extremely high VIF values, far exceeding the common threshold of 10. This level of multicollinearity can inflate standard errors and obscure the true effect of individual predictors, which explains the lack of statistically significant coefficients observed earlier.

We earlier observed the presence of outliers in the original dataset. The training dataset, derived from the original data, is expected to inherit outliers as well, so we drew box plots to detect any presence of outliers:





Based on the box plots, we observed the presence of outliers in several features. Specifically, x_2 shows an outlier at observation 7, while x_3 exhibits multiple outliers at observations 7, 14, 19, 20, 23, and 26. In the case of x_6 , an outlier is detected at observation 2, and x_8 has an outlier at observation 26. These outliers could potentially influence the model's performance and interpretability. Therefore, after addressing singularities in the dataset to ensure model identifiability, we refitted the model by excluding the identified outliers,

resulting in Model 1.3. This revised model was explored in the hope of observing improvements in fit or coefficient stability. The updated model summary is presented below.

```
Call:
lm(formula = y ~ x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_8 + x_9,
    data = train_data_removedOutliers)

Residuals:
    Min       1Q   Median       3Q      Max
-1.6543 -0.4275 -0.0358  0.5970  1.7735

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.077758  18.863744   0.110   0.9139
x_1         -0.124129   0.835945  -0.148   0.8841
x_2          4.451546   4.442579   1.002   0.3333
x_3          0.009551   0.009751   0.979   0.3440
x_4         -1.657373   2.685038  -0.617   0.5470
x_5          5.629507   4.050116   1.390   0.1862
x_6         -1.925880   2.191682  -0.879   0.3944
x_8        -21.929684  10.349872  -2.119   0.0525 .
x_9         -0.361654   0.285718  -1.266   0.2263
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.194 on 14 degrees of freedom
Multiple R-squared:  0.7364,    Adjusted R-squared:  0.5858
F-statistic:  4.89 on 8 and 14 DF,  p-value: 0.004848
```

Result 5: OLS Model Summary fitted on the train dataset after deleting outliers (Model - 1.3)

This model shows that none of the individual predictors are statistically significant, suggesting that multicollinearity is a more prominent issue than the presence of outliers. A common remedy for multicollinearity is applying regularisation techniques such as Lasso or Ridge regression. However, in our case, we observed a significant presence of outliers in the dataset, and both Lasso and Ridge regression are sensitive to outliers due to their penalty terms. Specifically, Ridge regression is particularly susceptible to distortion from outliers, which compromises both its assumptions and reliability. Moreover, removing the outliers would have resulted in a substantial data loss of approximately 26.92%, potentially compromising the model's generalizability.

Even when outliers were removed from the dataset, additional outliers continued to emerge (the dataset showed 2 new outliers when the initially detected 7 outliers were dropped from the training dataset), suggesting the data's inherent susceptibility to extreme values. Given these challenges, we decided not to proceed with regularisation techniques such as Lasso or Ridge regression. Instead, we opted to explore feature selection techniques (like: step-wise regression), which can help mitigate multicollinearity while maintaining robustness to outliers and preserving the interpretability and stability of the model.

ii) Step-wise Linear Regression (Both-ways):

As a remedy to this issue and as a feature selection technique, I am proceeding with stepwise regression using a two-direction (forward and backward) selection model to reduce multicollinearity and identify a more parsimonious and interpretable set of predictors. The bidirectional approach is particularly well-suited for this scenario because it allows for dynamic refinement of the model by iteratively adding and removing variables based on statistical criteria such as AIC, BIC, or p-values. This flexibility helps to prevent the inclusion of redundant or highly correlated predictors, which can distort coefficient estimates and inflate variance. Unlike pure forward or backward selection, the bidirectional method can reintroduce variables that become relevant in the presence of others, ensuring a more robust and adaptive model-building process. Given the presence of multicollinearity in the dataset and the need for effective dimensionality reduction, this hybrid stepwise method provides an optimal balance between complexity and explanatory power.

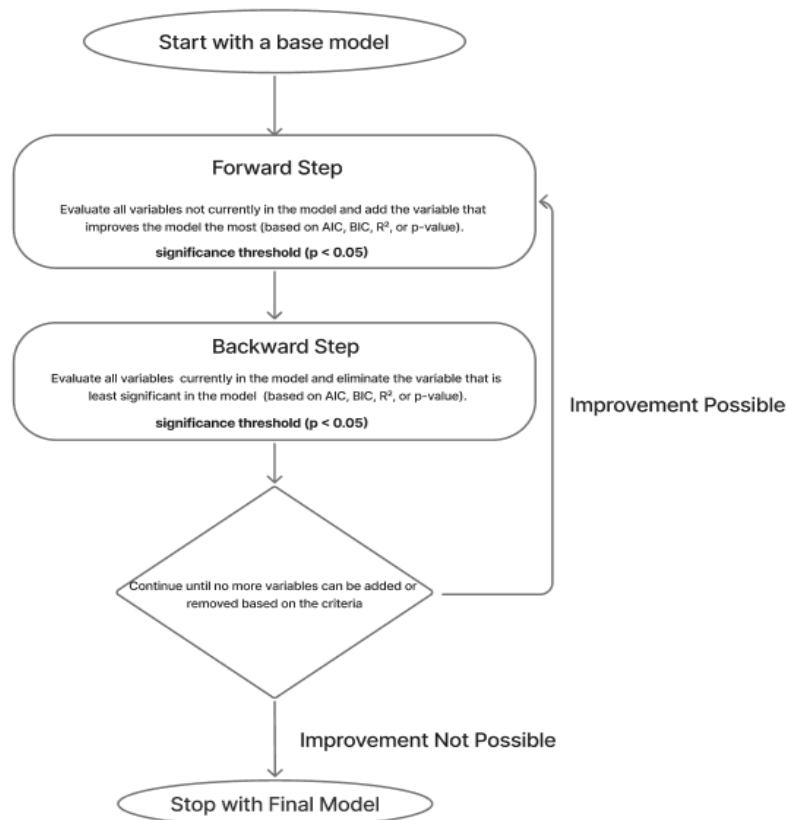


Diagram 5: Working mechanism of Bidirectional Step-wise Regression

The bothways stepwise regression model (Model: 2.1) on the training dataset returns the summary given below:

```

Residuals:
    Min       1Q   Median       3Q      Max
-1.92704 -0.48727 -0.07742  0.56145  2.35927

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -7.724561   9.883837  -0.782   0.4436
x_1          -0.768065   0.470100  -1.634   0.1179
x_2           5.336196   2.591570   2.059   0.0527 .
x_3           0.013476   0.007299   1.846   0.0797 .
x_5           1.202803   0.202741   5.933 8.4e-06 ***
x_8          -8.593010   3.762480  -2.284   0.0334 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.163 on 20 degrees of freedom
Multiple R-squared:  0.6947,    Adjusted R-squared:  0.6184
F-statistic: 9.101 on 5 and 20 DF,  p-value: 0.0001227

```

Result 6: OLS Model Summary fitted on the train dataset after deleting outliers (Model - 2.1)

We can see that the stepwise regression model (Model - 2.1) has improved interpretability by selecting the most influential predictors while maintaining a strong model fit. Variables x_5 and x_8 show strong statistical significance. Variables x_2 and x_3 may still hold some explanatory power. This model addresses the earlier multicollinearity issue, which was evident from high VIF scores in the full model. Therefore, **stepwise regression (both-way) remedied multicollinearity** and resulted in a cleaner, more interpretable model.

However, despite successfully mitigating multicollinearity, the residual standard error remains notably high at 1.163. This raises concerns about the model's fidelity to the underlying assumptions of Ordinary Least Squares (OLS) regression. In response, we examined the residuals versus fitted values plot to assess whether the model adheres to the assumption of homoscedasticity:

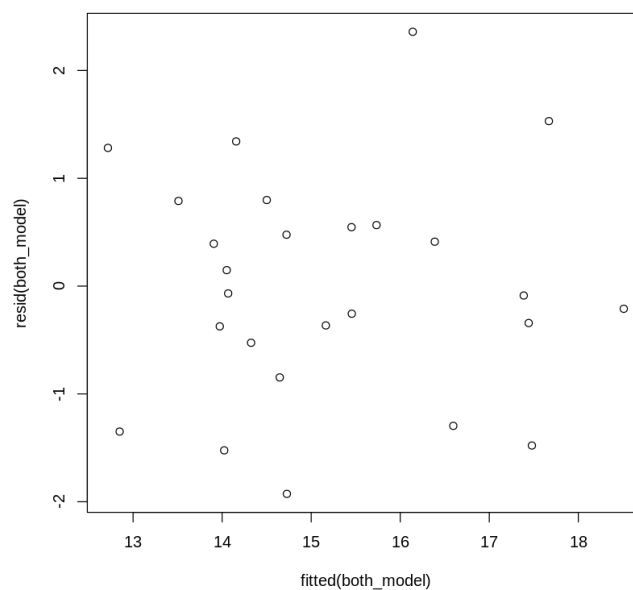


Diagram 6: Fitted vs Residual Plot for Model: 2.1

The resulting diagram presents a **fan-shaped pattern**, which is indicative of **heteroscedasticity**.

Following this finding, and considering the violations identified, namely heteroscedasticity, multicollinearity in the training dataset, and significant outlier influence, we now shift our focus to modelling approaches less reliant on strict assumptions. These alternative methods may offer more robust and reliable solutions for our data context.

iii) Robust Linear Regression Techniques:

To address these challenges, we explore robust linear regression techniques that are less dependent on stringent assumptions. These methods offer a more resilient modelling strategy under conditions where OLS fails to provide stable or interpretable results.

One such method is Robust Regression using *Huber's Loss Function*, which serves as the foundational approach in our assumption-free modelling setup.

- **M-estimator (Huber's Loss Function):** Introduced by Peter J. Huber in 1964 [\[7\]](#), Huber's loss function combines the squared loss (used in OLS) and the absolute loss (used in Least Absolute Deviation regression) to balance sensitivity and robustness. It behaves quadratically for small residuals and linearly for large residuals, thus retaining efficiency for well-behaved data while resisting the influence of outliers. This approach defines a specific type of **M-estimator** (Maximum likelihood-type estimator), where the choice of the loss function $\rho(r)$ is Huber's loss. Like OLS (which uses $\rho(r) = r^2$), Huber's method minimises a sum of ρ -transformed residuals, providing a robust alternative to least squares estimation.

Mathematically, the Huber loss function is defined as:

$$L_{\delta}(r) = \begin{cases} \frac{1}{2}r^2 & \text{for } |r| \leq \delta \\ \delta(|r| - \frac{1}{2}\delta) & \text{for } |r| > \delta \end{cases}$$

where, $r = y_i - \hat{y}_i$: residual;

δ : hyperparameter [tuning parameter that determines the cutoff between quadratic and linear behaviour]

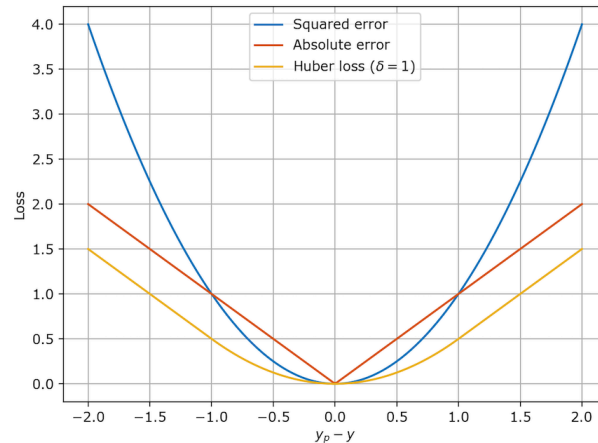


Diagram 7: Huber's Loss Function

Huber's regression emerges as a practical and principled alternative. It offers a more stable coefficient estimate, better predictive performance under noise, and robustness to data irregularities like outliers, heavy-tailed distributions (Non-Gaussian Error Distributions), and variable error spreads (Heteroscedastic Noise).

By replacing the MSE function with Huber's loss function in our initial model (Model: 1.2), a new model is obtained (Model: 3.1). The summary is given below:

```
Call: rlm(formula = y ~ x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_8 + x_9,
  data = train_data, psi = psi.huber)
Residuals:
    Min       1Q   Median       3Q      Max
-2.0143 -0.6068  0.1272  0.4721  2.2530

Coefficients:
              Value      Std. Error t value
(Intercept) -8.2084    19.4618   -0.4218
x_1          -0.9339     0.7788   -1.1992
x_2           6.1926     4.7243    1.3108
x_3           0.0149     0.0105    1.4258
x_4          -0.4763     2.8746   -0.1657
x_5           2.8197     4.0374    0.6984
x_6          -1.4506     2.4044   -0.6033
x_8          -14.5551     9.6415   -1.5096
x_9           -0.1134     0.2732   -0.4151

Residual standard error: 0.7845 on 17 degrees of freedom
```

$T_{(0.05,17)} = \pm 2.110$ [two - tailed]
 $T_{(0.01,17)} = \pm 2.898$ [two - tailed]

Result 7: Summary for Model 3.1

Although Huber's loss can deal with outliers and moderately handle heteroscedasticity, it's still susceptible to multicollinearity, which we can observe in Result 7, despite of **Residual standard error**: 0.7845 on 17 degrees of freedom — implying relatively low model error none of **the predictors are statistically significant** at typical significance levels (like 0.05), based on the t-values. Moreover, the model's R^2 (0.7133) and Adjusted R^2 (0.5783) indicate that while a fair proportion of variance is explained, the penalty for model complexity reflects potential overfitting or collinearity among predictors.

→ **MM Method:** The MM Method is a special type of M-estimation developed by Yohai [16], though not explicitly designed for dealing with multicollinearity, robust regression using the MM Method is generally preferred over Huber's loss when all three issues: multicollinearity, outliers, and heteroscedasticity, are present.

- ◆ Huber's technique has a moderate resistance to outliers but is still not fully robust to high-leverage outliers (especially in predictors). The MM Method is designed with a very high breakdown point (up to 50%), meaning it can tolerate a large proportion of outliers before the estimates become unreliable.
- ◆ The MM Method uses a combination of **S-estimators** (introduced by Rousseeuw and Yohai for robustness) [15] and **M-estimators** (for efficiency), making it **more flexible under non-constant variance** conditions than **Huber Loss**.
- ◆ **Robust initial estimates** used in the MM Method (like S-estimators) help avoid the instability that collinearity introduces in standard regression, which is absent in Huber's Loss technique.
- ◆ The MM Method applies **multi-stage reweighting instead of a single reweighting scheme for residuals like Huber's Loss technique**, first estimating a robust scale, then robust coefficients, and finally refining them, leading to **more accurate and stable estimates**

Yohai (1987) outlined a three-step procedure for constructing an MM-estimator, combining both high breakdown robustness and high statistical efficiency:

1. Begin with a robust initial estimate of the regression coefficients, denoted as $\hat{\beta}$, obtained using a high breakdown-point method. This preliminary estimator ensures resistance to outliers. Based on this, compute the residuals $r_i(\beta) = y_i - x_i^T \hat{\beta}$.
2. Using the residuals from Step 1, estimate the scale of the residuals with a function ρ that achieves a breakdown point of 50%. The scale estimator s_n is derived such that:

$$\frac{1}{n} \sum_{i=1}^n \rho\left(\frac{r_i}{s_n}\right) = k$$

where, k is a constant and s_n represents the robust estimate of the residual spread.

3. The final MM-estimator is derived by solving a system of estimating equations based on a redescending influence function $\varphi_1(u) = \frac{\delta \rho_1(u)}{\delta u}$, which uses the scale s_n obtained earlier. The estimate $\hat{\hat{\beta}}$ satisfies the equation:

$$\sum_{i=1}^n x_{ij} \varphi_1\left(\frac{y_i - \hat{x}_i \beta}{s_n}\right) = 0 \quad \text{for } j = 1(1)p$$

Now, as an improvement from Huber's Loss technique, the MM Method has been applied to generate a new model (Model 3.2) which will deal with all arising anomalies in the training dataset. By fitting the dataset into the model, the given result is obtained:

```
lmrob(formula = y ~ x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_8 + x_9, data = train_data,
      method = "MM")
\--> method = "MM"
Residuals:
    Min       1Q   Median       3Q      Max
-4.8231 -0.5730  0.1021  0.3658  2.2518

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -7.609163    9.845506  -0.773   0.4502
x_1          -0.581110    0.819926  -0.709   0.4881
x_2           4.617232    2.616750   1.764   0.0956 .
x_3           0.008216    0.009920   0.828   0.4190
x_4          -2.086893    1.959675  -1.065   0.3018
x_5           4.093641    2.234675   1.832   0.0845 .
x_6          -0.190132    1.629617  -0.117   0.9085
x_8           4.706333   12.711890   0.370   0.7158
x_9           0.080280    0.166883   0.481   0.6366
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Robust residual standard error: 1.003
Multiple R-squared:  0.7278,    Adjusted R-squared:  0.5997
Convergence in 18 IRWLS iterations
```

Result 8: Summary of Model 3.2

The MM Method model (Model 3.2) demonstrates a strong overall fit, with a Multiple R-squared value of 0.7278 and an Adjusted R-squared of 0.5997, indicating that approximately 60% of the variability in the response variable is explained after accounting for the number of predictors. This level of explanatory power is considered satisfactory in the current context. The robust residual standard error is also relatively low (1.003), suggesting the model has captured the underlying pattern in the data reasonably well, even in the presence of potential outliers.

Among the predictors, x_2 and x_5 show marginal statistical significance at the 10% level with positive coefficients, hinting at a potentially meaningful influence on the response variable. However, the coefficients of other predictors, including x_1 , x_3 , x_4 , x_6 , x_8 and x_9 , are not statistically significant, which may imply either a weak relationship or insufficient evidence to detect a true effect under the current sample conditions.

Model Fitting

After trying to fit different multiple regression models to identify the most suitable Model to predict the response variable y , in this section, we'll compare them to identify the best-fitted model (based on Adjusted R^2). The models differ in terms of estimation techniques and robustness to outliers or multicollinearity. Below are the models we tried, along with their respective Adjusted R^2 values:

1. **Model 1.2:** Ordinary Least Squares (OLS) Regression
 - Adjusted R^2 : 0.5883
2. **Model 1.3:** Ordinary Least Squares (OLS) Regression after dropping outliers
 - Adjusted R^2 : 0.5858
3. **Model 2.1:** Bothways Stepwise Linear Regression
 - Adjusted R^2 : 0.6184
4. **Model 3.1:** Robust Regression (using Huber's loss Function)
 - Adjusted R^2 : 0.5783
5. **Model 3.2:** Robust Regression (MM-estimation via `lmrob`)
 - Adjusted R^2 : 0.5997

Model Comparison

Among all the models, the Bothways Stepwise Linear Regression achieved the highest Adjusted R^2 of 0.6184, indicating the best performance in explaining the variability in the response variable while appropriately penalising for model complexity. However, this is also the model that revealed a major violation of the homoscedasticity assumption in the dataset, which raises concerns about the validity of its estimates.

To address this issue, we explored robust algorithms that are more resilient to such violations. Notably, the Robust Regression (MM-estimation) performed well despite the presence of potential outliers, achieving a respectable Adjusted R^2 of 0.5997. This demonstrates its strength in providing stable and reliable estimates under non-ideal data conditions.

Based on this comparison, the Robust Regression (MM-estimation via `lmrob`) is selected as the most suitable model for further analysis and interpretation. In contrast, the other models—including Ordinary Least Squares (with and without outliers) and Robust Regression using Huber's loss—yielded lower Adjusted R^2 values, suggesting issues like sensitivity to outliers or lower predictive power.

So the best fitted model is:

$$y = -7.6092 - 0.5811x_1 + 4.6172x_2 + 0.0082x_3 - 2.0868x_4 + 4.0936x_5 \\ - 0.1901x_6 + 4.7036x_8 + 0.082x_9$$

Cross-validation

Following the model fitting stage, the next critical step is to evaluate the generalizability and predictive performance of the developed models. While a model may perform well on the training data, it is essential to assess how accurately it can predict outcomes on unseen data. This is where cross-validation plays a key role. In this section, we conduct a comprehensive comparison by applying the previously derived models exclusively to a separate test dataset, without involving the training data. The objective is to identify the model that best balances goodness of fit with strong predictive power in real-world scenarios. Metrics such as Root Mean Squared Error (RMSE) on the test data are used to evaluate model performance. This validation procedure allows us to determine which model serves as the most reliable predictor, offering robust and generalizable results. The comparative results of the models' output and performance on the test data are presented below.

Output:

y	Predicted_y [By Model: 1.2]	Predicted_y [By Model: 1.3]	Predicted_y [By Model: 2.1]	Predicted_y [By Model: 3.1]	Predicted_y [By Model: 3.2]
17.3	19.58356	20.03193	20.46131	19.51955	18.40308
16.5	17.78401	17.41289	17.83387	17.69552	17.47262
15.8	15.09868	14.01295	14.95184	15.32609	14.13033
16.3	15.5939	15.56706	15.50387	15.48572	15.79535
15.7	16.8737	17.09276	16.39938	16.6757	17.26971
14	15.76292	14.95171	16.18565	15.79256	15.28372

Performance as a Predictor Model:

Model No.	Model Type	RMSE
Model: 1.2	OLS Model	1.434063
Model: 1.3	OLS Model after dropping outliers	1.574437
Model: 2.1	Bothway Stepwise Regression Model	1.750833
Model: 3.1	Robust Regression with M estimator (Huber's loss function)	1.378922
Model: 3.2	Robust Regression with MM estimator	1.246141

From the above results, it's proven that for our dataset, Robust Regression with MM estimator is the best predictor model among others. So we can conclude that Model 3.2 is the best predictor.

Conclusions

In this project, we explored the challenge of predicting wine quality using a small, multicollinear, and outlier-affected dataset. Starting with Ordinary Least Squares (OLS) regression as a baseline, we uncovered limitations stemming from multicollinearity and sensitivity to outliers, which compromised the stability and interpretability of the model. Through stepwise selection, we improved model parsimony, but residual diagnostics revealed persistent issues such as heteroscedasticity.

To overcome these challenges, we turned to robust regression techniques, particularly Huber's M-estimator and the MM-estimator. Among the approaches compared, the MM-estimator emerged as the most effective method, achieving the best trade-off between predictive accuracy and resistance to data irregularities as demonstrated by the lowest RMSE and a relatively high Adjusted R^2 . This robust method proved especially valuable in handling small-sample data with violations of key linear regression assumptions.

The findings underscore the importance of choosing adaptive and assumption-resilient models in practical data science applications. Particularly in domains like oenology, where quality assessment can benefit from more objective, data-driven methods, such approaches offer a reliable framework for supporting decision-making and quality control.

While the dataset size and lack of metadata presented certain limitations, the modelling pipeline developed in this study can serve as a generalisable blueprint for future studies in wine quality assessment or similar regression problems involving complex real-world data. Further research could explore advanced regularisation techniques like Robust Ridge Regression, expand the dataset for improved statistical power and improve generalisability without sacrificing interpretability.

Limitations and Future Scope

One of the most critical limitations of this study is the small dataset size ($n = 32$), which restricts the applicability of data-hungry machine learning methods such as neural networks, ensemble learning, or deep regression models. The limited sample also reduces the statistical power of classical regression models, leading to inflated standard errors and low significance levels across several predictors, particularly in Model 3.2. While the MM Method showed robustness against outliers and heteroscedasticity, it does not inherently resolve multicollinearity issues.

Additionally, although various regression techniques were explored, dimensionality reduction methods such as **Principal Component Analysis (PCA)** or **Partial Least Squares (PLS)** could have been considered either as alternative modelling frameworks or as feature selection strategies to mitigate multicollinearity while preserving interpretability.

Another area for methodological enhancement involves **model validation**. The current analysis employed a single 80:20 train-test split, which can lead to variance in performance evaluation depending on the partition. Implementing **k-fold cross-validation** would allow a more reliable and generalizable assessment of model performance across different data splits.

Although I had initially considered extending the study using **Robust Ridge Regression** to further address the multicollinearity issue in Model 3.2, this direction was not pursued due to time constraints and limited familiarity with the technique at the time of execution. Nevertheless, it remains a viable and promising path for future work.

In summary, expanding the dataset, integrating dimensionality reduction or regularisation-based approaches, and employing robust validation methods like k-fold cross-validation can significantly strengthen both the statistical and practical value of the findings. These steps may help achieve a better balance between **model robustness, explanatory power, and statistical significance** in future studies.

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Appendix

Train Dataset [Dimension: 26 × 11]:

y	x₁	x₂	x₃	x₄	x₅	x₆	x₇	x₈	x₉	x₁₀
19.2	0	3.85	66	9.35	5.65	2.4	3.25	0.33	19	0.065
18.3	0	3.73	79	11.15	6.95	3.15	3.8	0.36	21	0.076
17.1	0	3.88	73	9.4	5.75	2.1	3.65	0.4	18	0.073
16.8	0	3.98	75	8.55	5.05	2.05	3	0.49	12	0.06
15.2	0	3.66	86	6.4	4	1.5	2.5	0.27	19	0.05
15.2	0	3.91	78	5.8	3.3	1.4	1.9	0.4	9	0.038
14	0	3.47	178	3.6	2.25	0.75	1.5	0.37	8	0.03
14	0	3.91	81	3.9	2.15	1	1.15	0.32	7	0.023
13.8	0	3.75	108	5.8	3.2	1.6	1.6	0.38	8	0.032
13.6	0	3.9	92	5.4	2.85	1.55	1.3	0.44	6	0.026
12.8	0	3.92	96	5	2.7	1.4	1.3	0.35	7	0.026
18.5	1	3.87	89	9.15	5.6	1.95	3.65	0.46	16	0.073
17.3	1	3.97	59	10.25	6.1	2.4	3.7	0.4	19	0.074
16.3	1	3.76	22	8.2	5	1.85	3.15	0.25	25	0.063
16	1	3.98	58	10.15	6	2.6	3.4	0.38	18	0.068
16	1	3.88	85	6.85	4.1	1.5	2.6	0.33	16	0.052
15.5	1	3.98	94	5.45	3.05	1.5	1.55	0.41	8	0.031
15.3	1	3.69	122	8	5.05	1.9	3.15	0.27	23	0.063
15.3	1	3.77	144	5.6	3.35	1.1	2.25	0.36	12	0.045
14.8	1	3.74	10	7.9	4.75	1.95	2.8	0.25	23	0.056
14.3	1	3.76	100	5.55	3.25	1.15	2.1	0.34	12	0.042
14.3	1	3.91	73	4.65	2.7	0.95	1.75	0.36	10	0.035
14.2	1	3.6	301	4.25	2.4	1.25	1.15	0.42	6	0.023
13.8	1	3.9	67	7.4	4.4	1.6	2.8	0.45	13	0.056
12.5	1	3.8	89	5.35	3.15	1.2	1.95	0.32	12	0.039
11.5	1	3.65	192	6.35	3.9	1.25	2.65	0.63	8	0.053

Test Dataset [Dimension: 6 × 11]

y	x₁	x₂	x₃	x₄	x₅	x₆	x₇	x₈	x₉	x₁₀
17.3	0	3.86	99	12.85	7.7	3.9	3.8	0.35	22	0.076
16.5	0	3.85	61	10.3	6.2	2.5	3.7	0.38	20	0.074
15.8	0	3.93	66	4.9	2.75	1.2	1.55	0.29	11	0.031
16.3	1	3.76	77	8.35	5.05	1.9	3.15	0.37	17	0.063
15.7	1	3.75	120	8.8	5.5	1.85	3.65	0.39	19	0.073
14	1	3.76	104	8.7	5.1	2.25	2.85	0.34	17	0.057

R Code: ²

Install libraries:

```
install.packages("car")
install.packages("corrplot")
install.packages("glmnet")
install.packages("robust")
```

Loading libraries:

```
library(MASS) # For robust regression
library(car)  # For diagnostic checks
library(robustbase)
library(corrplot)
library(glmnet)
```

Loading a public CSV file using ID and storing it into a variable:

```
system("gdown --id 1pu_RYc-NFbGCGLtpFydX_R0_DHROc6kn")
system("ls", TRUE)

DATA <- read.csv("WineProduction_SNU_Project_SemVI.csv", header = TRUE)
```

Printing the data head

```
head(DATA)
any(is.na(DATA))
```

Performing EDA

```
cv <- sapply(DATA, function(x) if(is.numeric(x)) sd(x, na.rm =
TRUE)/mean(x, na.rm = TRUE) * 100 else NA)
print(cv)
```

Plotting scatter plots

```
par(mfrow = c(3, 4))
for (col in colnames(DATA)) {
  if (col != "y" && is.numeric(DATA[[col]])) { # Ensure it's numeric
    plot(DATA[[col]], DATA$y,
         main = paste("Scatter Plot of y vs ", col),
         xlab = col, ylab = "y",
         col = "black", pch = 19)
  }
}
par(mfrow = c(1, 1))
```

² Use this link to access the code: [Click Here](#) [Access is subjective upon approval]

Drawing histograms

```
par(mfrow = c(1,1))
for (col in colnames(DATA)) {
  hist(train_data[[col]],
       main = paste("Histogram of", col),
       xlab = col,
       col = "blue",
       border = "yellow")
}
par(mfrow = c(1, 1))
```

Plotting Box Plots

```
par(mfrow = c(1, 1)) # Adjust based on the number of columns

# Create an empty list to store outlier indices for each column
outlier_list <- list()

# Loop through each column in train_data
for (col in colnames(DATA)) {
  # Check if the column is numeric before plotting
  if (is.numeric(DATA[[col]])) {

    column_data <- DATA[[col]]

    # Identify outliers using boxplot.stats
    outliers <- boxplot.stats(column_data)$out

    # Save indices of outliers
    if (length(outliers) > 0) {
      outlier_indices <- which(column_data %in% outliers)
      outlier_list[[col]] <- outlier_indices # Store by column name
    }

    # Draw boxplot
    boxplot(column_data, main = paste("Boxplot of", col),
            col = "light green", ylab = col, border = "darkblue")
  }
}

# Reset plot layout to default
par(mfrow = c(1, 1))
```



```
# Print outlier indices list
print(outlier_list)
```

Correlation Matrix & Corrplot

```
Cor1 = cor(DATA)
Cor1
corrplot(Cor1)
```

Splitting the dataset into train and test

```
set.seed(1)
sample <- sample(c(TRUE, FALSE), nrow(DATA), replace=TRUE, prob=c(0.8,
0.2))
train_data <- DATA[sample, ]
test_data <- DATA[!sample, ]
cat("Train set dimensions:", dim(train_data), "\n")
cat("Test set dimensions:", dim(test_data), "\n")
print(train_data)
print(length(train_data))
print(test_data)
```

Copy of a test dataset

```
test_data_precited <- test_data
```

Simple OLS Model and Summary

```
ols_model <- lm(y ~ x_1+x_2+x_3+x_4+x_5+x_6+x_7+x_8+x_9+x_10, data =
train_data)
summary(ols_model)
```

Plotting Simple OLS

```
par(mfrow = c(1,1))
plot(ols_model)
```

Checking for alias

```
alias(lm(y ~ ., data = train_data))
```

Remove linearly dependent variables and fit into OLS

```
train_data_cleaned <- train_data[, !apply(train_data, 2, function(x)
all(x == x[1]))]
#print(train_data_cleaned)
ols_cleaned_model = lm(y ~ x_1+x_2+x_3+x_4+x_5+x_6+x_8+x_9, data =
train_data_cleaned)
```

```
summary(ols_cleaned_model)
```

Checking VIF Scores

```
vif_values <- vif(ols_cleaned_model) # Use OLS to check VIF
print(vif_values)
```

Detecting outliers in the Train Dataset

```
par(mfrow = c(1, 1)) # Adjust based on the number of columns

# Create an empty list to store outlier indices for each column
outlier_list <- list()

# Loop through each column in train_data
for (col in colnames(train_data)) {
  # Check if the column is numeric before plotting
  if (is.numeric(train_data[[col]])) {

    column_data <- train_data[[col]]

    # Identify outliers using boxplot.stats
    outliers <- boxplot.stats(column_data)$out

    # Save indices of outliers
    if (length(outliers) > 0) {
      outlier_indices <- which(column_data %in% outliers)
      outlier_list[[col]] <- outlier_indices # Store by column name
    }

    # Draw boxplot
    boxplot(column_data, main = paste("Boxplot of", col),
            col = "light blue", ylab = col, border = "darkblue")
  }
}

# Reset plot layout to default
par(mfrow = c(1, 1))

# Print outlier indices list
print(outlier_list)
```

Fitting into OLS after dropping outliers

```
ols_outliersDropped_model = lm(y ~
x_1+x_2+x_3+x_4+x_5+x_6+x_7+x_8+x_9+x_10, data =
train_data_removedOutliers)
summary(ols_outliersDropped_model)
ols_outliersDropped_model2 = lm(y ~ x_1+x_2+x_3+x_4+x_5+x_6+x_8+x_9,
data = train_data_removedOutliers)
summary(ols_outliersDropped_model2)
```

Checking for VIF Score

```
vif_values_withoutOutliers <- vif(ols_outliersDropped_model2) # Use
OLS to check VIF
print(vif_values_withoutOutliers)
```

Fitting into Both-way Stepwise Regression Model and Summarising

```
both_model <- step(ols_model, direction = "both", trace = 0)
summary(both_model)
```

Checking VIF Scores for “both_model”

```
vif_scores <- vif(both_model)
print(vif_scores)
```

Plotting Fitted vs. Residual Plot

```
plot(fitted(both_model), resid(both_model))
```

Robust regression Model using Huber's loss

```
robust_model <- rlm(y ~ x_1+x_2+x_3+x_4+x_5+x_6+x_8+x_9, data =
train_data, psi = psi.huber)

# Summary of robust model

summary(robust_model)
```

Calculating R^2 and Adjusted R^2 :

```
get_adj_r2_rlm <- function(model, x, y) {
  y_hat <- predict(model)
  ss_res <- sum((y - y_hat)^2)
  ss_tot <- sum((y - mean(y))^2)
  r2 <- 1 - ss_res / ss_tot
  n <- length(y)
  p <- length(coef(model)) - 1
  adj_r2 <- 1 - ((1 - r2) * (n - 1)) / (n - p - 1)
  return(c(R2 = r2, Adjusted_R2 = adj_r2))
}
```

```
get_adj_r2_rlm(robust_model, x, y)
```

Checkin VIF Scores:

```
vif_robust_scores <- vif(robust_model)
print(vif_robust_scores)
```

Robust regression Model based on MM Method

```
robust_model_singular <- lmrob(y ~
x_1+x_2+x_3+x_4+x_5+x_6+x_7+x_8+x_9+x_10, data = train_data, method =
"MM")
summary(robust_model_singular)
```

Prediction on Test Data:

```
#Prediction based on OLS MODEL (After removing singularities)
test_data_precited$predicted_y_olsModel2 <- predict(ols_cleaned_model,
newdata = test_data)
#Prediction based on OLS MODEL (After dropping outlier)
test_data_precited$predicted_y_OutlierDroppedModel<-
predict(ols_outliersDropped_model2, newdata = test_data)
#Prediction based on Bothway Stepwise Model
test_data_precited$predicted_y_bothModel <- predict(both_model, newdata
= test_data)
#Prediction based on Robust Model (Using Huber's Loss)
test_data_precited$predicted_y_robust_huber <- predict(robust_model,
newdata = test_data)
#Prediction based on robust Model (Using MM Method)
test_data_precited$predicted_y_robust_MM <- predict(robust_model2,
newdata = test_data)
```

Printing test data with all predictions:

```
print(test_data)
print(test_data_precited)
```

Model Evaluation [will print RMSE and R^2 for each model]

```
actuals <- test_data_precited$y
# List of prediction column names
prediction_columns <- c(
  "predicted_y_olsModel2",
  "predicted_y_OutlierDroppedModel",
  "predicted_y_bothModel",
```

```

    "predicted_y_robust_huber",
    "predicted_y_robust_MM"
  )
  # Function to compute RMSE and R²
  evaluate_model <- function(pred_column) {
    predicted <- test_data_predicted[[pred_column]]
    rmse <- sqrt(mean((actuals - predicted)^2))
    r_squared <- cor(actuals, predicted)^2
    return(c(RMSE = rmse, R_squared = r_squared))
  }

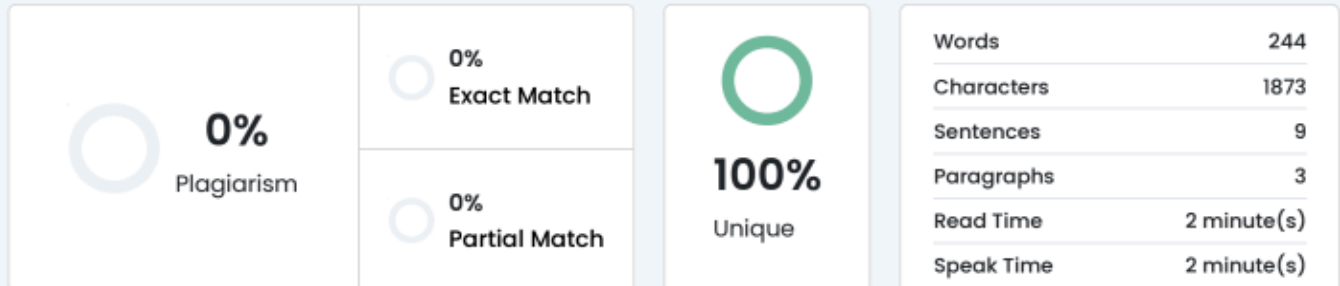
  # Apply the function to each model
  results <- sapply(prediction_columns, evaluate_model)

  # Transpose for readability (models as rows)
  results <- t(results)
  print(results)

```

Plagiarism Report

Plagiarism Scan Report



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Abstract

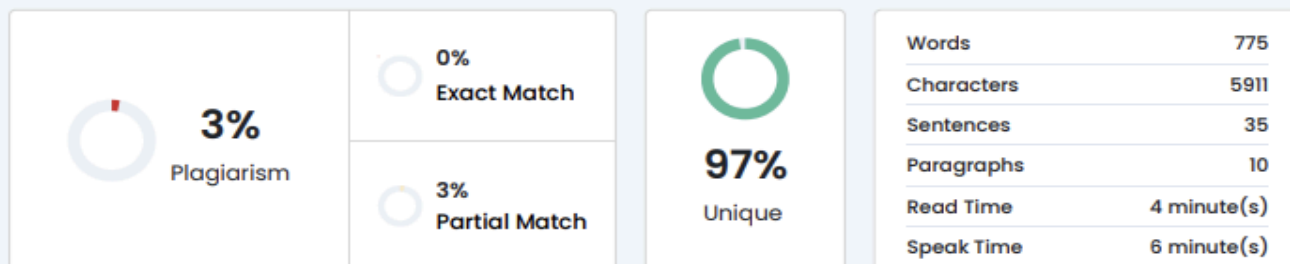
This project explores the application of various linear regression techniques for predicting wine quality, which has a very important role, particularly in the wine industry for shaping up the preferences of consumers, influencing the pricing related strategies and guiding decision making in production, using a dataset characterised by multicollinearity and a significant presence of outliers. The dataset comprises 32 wine samples, each evaluated across 10 physicochemical attributes - including pH, total sulphur dioxide, anthocyanin concentration and colour density - with quality scores assigned on a 0-20 scale. An extensive Exploratory Data Analysis (EDA) revealed distributional skewness, strong inter-variable correlations, and a high incidence of outliers. The initial application of Ordinary Least Squares (OLS) regression exposed limitations due to multicollinearity, as confirmed through elevated Variance Inflation Factors (VIFs). Stepwise regression (both-way) improved model parsimony, but heteroscedasticity and sensitivity to outliers persisted. To address these, robust regression approaches were adopted, including Huber's M-estimator and the MM-estimator. Comparative analysis using metrics such as Adjusted R^2 and Root Mean Square Error (RMSE) demonstrated the MM-estimator's superior resilience to data irregularities. Ultimately, the MM-estimator emerged as the most reliable and interpretable model, offering a robust framework for data-driven wine quality assessment and decision-making in viticulture. This project thus underscores the importance of robust techniques in real-world data environments and presents a generalisable modelling framework to support objective wine quality assessment in the wine industry.

Keywords: Wine quality rating; Exploratory Data Analysis; Ordinary Least Squares; Stepwise Regression; Robust Regression

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Introduction

Wine quality plays a crucial role in the wine industry, shaping consumer preferences, influencing pricing strategies, and guiding production decisions. Traditionally, quality assessment has relied on sensory evaluations conducted by expert sommeliers. While valuable, such evaluations can be subjective and prone to inconsistency. The perceived quality of wine is influenced by various physicochemical properties, including acidity, sulfur dioxide content, colour density, and anthocyanin composition. Understanding the quantitative relationship between these measurable features and wine quality allows for a more objective and systematic evaluation process, reducing dependence on human judgment. With the rise of data-driven approaches in oenology, there is growing interest in predictive modelling techniques that assess wine quality based on these measurable characteristics.

Numerous projects have explored the feasibility of using statistical and machine learning methods for wine quality prediction. For instance, Dahal et al. (2021) evaluated models such as Ridge Regression, Support Vector Machines, and Gradient Boosting, demonstrating improved performance over traditional techniques, especially in handling high-dimensional data [1]. Similarly, Arshad (2024) employed ensemble models like Random Forest and XGBoost, reporting over 95% accuracy on larger, well-curated datasets [2]. However, these projects often overlook the effects of multicollinearity and outliers—issues that can significantly degrade model performance in practical settings.

Addressing this gap, the present project investigates a modest yet noisy dataset comprising 32 wine samples, each described by 10 physicochemical variables, including pH, total sulphur dioxide, anthocyanin concentration, and colour density. The response variable is a wine quality score on a 0–20 scale, derived from sensory assessments. The dataset presents two key challenges: multicollinearity among predictors and the presence of outliers, necessitating the adoption of robust, adaptive modelling strategies suited to real-world data.

Before applying regression techniques, a comprehensive Exploratory Data Analysis (EDA) is conducted to understand the structure and variability of the data. Techniques such as histograms and density plots are used to assess the skewness, modality, and distribution of individual variables, helping to identify those requiring transformation. Boxplots assist in outlier detection, while scatterplots and correlation matrices are used to examine bivariate relationships and detect multicollinearity. Insights from the EDA inform the selection of appropriate modelling techniques, ensuring that the subsequent analysis is both accurate and resilient to data irregularities.

The modelling phase of this project begins with consideration of the foundational assumptions underlying linear regression models—namely, linearity, independence of residuals, and homoscedasticity. Although these assumptions are not formally tested prior to modelling, they are assessed contextually using insights derived from EDA and residual diagnostics throughout the modelling process. The initial modelling approach employs the Ordinary Least Squares (OLS) method, providing a baseline for performance and interpretability. However, the presence of multicollinearity among predictors could necessitate the exploration of variable selection techniques to improve model robustness.

To mitigate multicollinearity, we might apply stepwise regression techniques, including Forward Selection, Backward Elimination, and Bidirectional Stepwise Selection. These methods iteratively add or remove predictors based on criteria such as the Akaike Information Criterion (AIC), allowing the model to retain only those variables that meaningfully contribute to predictive accuracy. Despite the common use of Ridge Regression to handle multicollinearity, this technique is excluded from consideration due to its sensitivity to outliers. Ridge Regression minimises the sum of squared residuals augmented by an L2 penalty term ($\lambda ||\beta||_2^2$), which disproportionately penalises large coefficients. In the presence of extreme data points, this can distort model estimates and reduce generalizability.

With over 25% of the data points in our small dataset identified as outliers – an insight revealed during EDA – it is neither statistically advisable nor feasible to remove them. Instead, the project adopts and derives the best robust regression techniques designed to accommodate such irregularities without compromising model integrity. These methods offer enhanced resistance to the influence of outliers, ensuring that model estimates remain stable and reliable even under data contamination.

By evaluating the predictive accuracy and interpretability of different regression approaches, this project seeks to identify a robust and generalizable model for wine quality prediction. The findings offer practical insights into the key physicochemical factors influencing perceived quality, contributing both to data-driven winemaking practices and the broader methodological discourse in predictive modelling.

The industrial significance of this project lies in its potential to support quality assurance and product development within the wine industry. Wine quality assessment, often reliant on expert tasters, can benefit from objective, data-driven models that standardise evaluation and reduce subjectivity. By identifying the key physicochemical attributes that influence wine quality, producers can fine-tune cultivation practices, fermentation processes, and blending strategies to consistently meet consumer expectations. Furthermore, predictive models can streamline quality control workflows, reduce production costs, and enhance competitiveness in global markets where quality differentiation plays a pivotal role in brand positioning.

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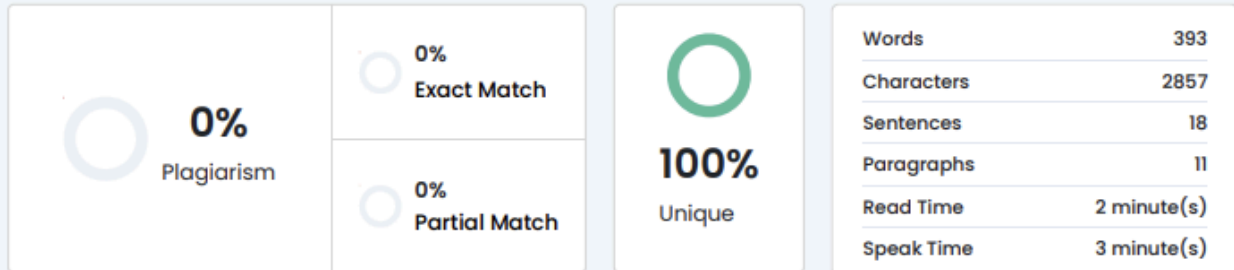
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Objectives

Understand Influencing Factors:

This project aims to uncover how various physicochemical properties, such as acidity, sugar content, pH levels, and alcohol concentration, influence the overall quality of wine. By identifying the most impactful features, the project will help highlight which aspects of wine production require more attention to improve quality outcomes.

EDA:

In the EDA phase, we will perform a comprehensive analysis using various visual and statistical techniques to understand the distribution, variability, and relationships among the variables. Scatter plots will be used to reveal potential linear and non-linear relationships between predictors and the target variable (wine quality), while histograms will examine the distribution and skewness of individual features. Box plots will help to detect outliers and compare feature distributions across different wine quality levels. Generating a correlation matrix will help to identify multicollinearity among features, highlighting which variables are strongly interrelated. Finally, the Coefficient of Variation (CV) will be calculated for each feature to assess their relative variability and importance in influencing wine quality. These insights will lay the foundation for feature selection and modelling decisions.

Build and Analyse Predictive Models:

The project will employ multiple regression-based linear modelling techniques to predict wine quality scores with a focus on accuracy, interpretability, and robustness. Techniques such as Ordinary Least Squares (OLS) regression, Stepwise regression, Regularisation Techniques and Robust regression (using Huber Loss, MM-estimator, etc.) will be explored as per the requirement of the analysis. These methods will allow for a comprehensive comparison of how different modelling strategies handle issues like multicollinearity, overfitting, the presence of outliers and others. The role of regularisation and variable selection will be particularly emphasised in enhancing model generalizability and stability under real-world data conditions.

Evaluate Model Performance:

To ensure reliability, the models will be evaluated using performance metrics such as Root Mean Squared Error (RMSE) and adjusted R-squared. These metrics will help assess how well each model fits the training dataset and how effectively it generalises to unseen wine data. This dual evaluation provides a balanced understanding of both the in-sample performance and the model's predictive power on new data, guiding the selection of the most suitable modelling approach.

Support Quality Control in Wine Production:

By predicting wine quality based on measurable chemical inputs, the study intends to support wine producers in maintaining consistent quality standards. The results could aid in data-driven decision-making in the wine industry.

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